Type MS · Interpret antibacterial culture investigations

Answer 1 of 5 · Question 8 part b.ii · 3 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b.ii in this mark scheme extract

Number					
8	(a)	(i)	A = flagellum; B = cell wall/capsule; C = nucleic acid/genetic material/bacterial chromosome/(loop of) DNA;		3 (grad)
		(ii)	pathogen;		1 (cler)
	(b)	(i)	• produced by microorganism/bacteria/fungi; • kills bacteria;		2 (grad)
		(ii)	A *no mark for giving X* • because larger clear zone; • antibiotic diffused from filter paper; • bacteria have been killed;		3 (exp)
		(iii)	• filter paper without antibiotic; • bacterial growth unaffected/no clear zones; • if antibiotic is causing effect;		3 (exp)

Total 12 marks

Type MS · Interpret antibacterial culture investigations

Answer 1 of 5 · Question 8 part b.ii · 3 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b.ii in this mark scheme extract

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Type MS · Interpret antibacterial culture investigations

Answer 2 of 5 · Question 7 part b.ii · 2 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

number				
7				
(a)	(i)	C; (pathogen) A not all bacteria cause disease B not all fungi cause disease D not all protozoa cause disease		1
	(ii)	• Ebola caused by a virus; • antibiotics only affect bacteria/not active against viruses;		2
(b)	(i)	• 700 - 180 = 520; • 520 ÷ 5 (= 104); • 100 (AU per hour);	Ecf max 2 marks if first mp incorrect but value divided by 5 to arrive at correct answer Max 1 if first answer correct but division is incorrect	3
	(ii)	• B is bactericidal; • because population decreases/kills bacteria;		2
	(iii)	• antibiotic B; • because it kills bacteria/causes population to decrease; • but symptoms may be less severe/mild/reduced; • antibiotic A leaves large numbers alive/bacterial population remains constant/only stops/reduces population growth; • these can still cause disease;		3 Total 11

Type MS · Interpret antibacterial culture investigations

Answer 2 of 5 · Question 7 part b.ii · 2 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Interpret antibacterial culture investigations

Answer 3 of 5 · Question 7 part b.iii · 3 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part b.iii in this mark scheme extract

number				
7				
(a)	(i)	C; (pathogen) A not all bacteria cause disease B not all fungi cause disease D not all protozoa cause disease		1
	(ii)	• Ebola caused by a virus; • antibiotics only affect bacteria/not active against viruses;		2
(b)	(i)	• 700 - 180 = 520; • 520 ÷ 5 (= 104); • 100 (AU per hour);	Ecf max 2 marks if first mp incorrect but value divided by 5 to arrive at correct answer Max 1 if first answer correct but division is incorrect	3
	(ii)	• B is bactericidal; • because population decreases/kills bacteria;		2
	(iii)	• antibiotic B; • because it kills bacteria/causes population to decrease; • but symptoms may be less severe/mild/reduced; • antibiotic A leaves large numbers alive/bacterial population remains constant/only stops/reduces population growth; • these can still cause disease;		3 Total 11

Type MS · Interpret antibacterial culture investigations

Answer 3 of 5 · Question 7 part b.iii · 3 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Interpret antibacterial culture investigations

Answer 4 of 5 · Question 5 part a.iii · 2 marks

2024 JUN · Paper 02 · find part a.iii in this mark scheme extract

Number				
5	(a) (i)	**C** the type of antibiotic added The following are incorrect because: A the mass of bacteria added is irrelevant B the size of the inhibition zone is the dependent variable D the mass of agar in the Petri dish is irrelevant		(1)
	(ii)	Any two from: • sterilise equipment/named equipment/disinfect/swab desk; • use of sterile agar; • reference to use of lid on Petri dish/ minimising the time the lid is removed; • Flame from Bunsen burner:		(2)
	(iii)	(antibiotic) Q; greater inhibition zone /greater clear zone/ more bacteria killed;		(2)
	(iv)	Any four from: • mutation; • (resistant) bacteria survive; • reproduce; • pass on resistant gene/allele; • Increase of resistant gene/allele in the population/ increase in allele frequency (over many generations)		(4)
	(b)	• (bacteria/fungi/decomposers) release enzymes/named enzyme • breakdown dead/organic matter/ named organic matter; • release nutrients/named nutrients into the soil/environment;	Ignore carbon dioxide Ignore fertiliser	(2)

Total for Question 5 = 11 marks

Type MS · Interpret antibacterial culture investigations

Answer 5 of 5 · Question 7 part b.ii · 4 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

number				
7 (a)	(i)	• temperature of incubation; • volume of milk; • composition of nutrients in agar/type of agar;		3
	(ii)	size of bacterial growth;		1
(b)	(i)	$\frac{7.1}{5} = 1.42$; $\frac{36.5}{5} = 7.3$;	One mark for each correct answer	2
	(ii)	• more bacteria in B; • at start of investigation; • because extra added to tube; • some bacteria in A/milk which grew;	Allow more growth in B/growth increases more quickly in B Allow reference to contamination e.g.from air	4
(c)		• transfer loop sterilised/heated in flame; • transfer performed quickly;	Allow reference to sterilised in context / sterile technique / sterilise equipment / named equipment	2
(d)		• repeat investigation; • using other organisms;		2

Total 14 marks

Type MS · Interpret antibacterial culture investigations

Answer 5 of 5 · Question 7 part b.ii · 4 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks